

Quantum-Enhanced Credit Card Fraud Detection for Digital Payment Ecosystems

Phase 1 concept proposal. Problem statement: HSBC, financial services / QML

Claude Shannon’s Fraud Catchers | September 2026

Summary. We built the apparatus HSBC’s question needs and ran it on real hardware. A tuned classical detector, a quantum-inspired candidate (CVQBoost with its training step solved on QCi’s Dirac-3), and an exact classical solve of the identical objective were evaluated under a preregistration frozen before any result was seen. The confirmatory endpoint returned a null: the quantum-inspired arm trails the matched classical detector by 0.0399 AUPRC on the ULB benchmark, on nine of ten seeds. The diagnosis is precise and measured: the continuous CVQBoost objective is convex, the device reproduces the classical optimum to within noise, and the value in this formulation sits in the weak learners, not the optimizer. One result is positive at scale: the 833-variable arm using three-feature learners gains +0.0256 AUPRC on ten of ten seeds, which points at exactly the cross-feature headroom HSBC’s statement names. Phase 2 tests the one formulation where a quantum solver would have to do real work, cardinality-constrained learner selection, against a certified classical optimum, and settles the temporal question a bank must answer before any production trial. The title above is HSBC’s name for the problem statement, kept for identification. We do not claim quantum enhancement or quantum advantage.

1. Problem framing

Card fraud detection is a decision under two hard constraints: a fixed analyst review budget and an asymmetric cost, where a miss is a chargeback and a false alarm is a lost customer. At the fraud rates of real portfolios (0.17% in the ULB benchmark, 3.5% in IEEE-CIS), accuracy is meaningless and ranking metrics matter only where the review budget binds, at the top of the list.

HSBC’s statement says classical models are strong (AUC-ROC 0.9459 best-in-class on IEEE-CIS) and that the headroom lies in subtler cross-feature patterns. We therefore ask a narrow, answerable question: can a quantum or quantum-inspired training step beat a tuned gradient-boosted ensemble on the metric a fraud team uses, fraud caught within a fixed review budget, under a protocol that cannot be moved after the result is seen? Phase 1 answers it in both directions and identifies the one experiment that can change the answer.

2. Technical approach

Paradigm: hybrid classical-quantum. Weak-learner construction and all inference are classical. Dirac-3 performs one photonic optimization step at training time. An exact classical convex solve of the identical objective runs as a structural control on every configuration. QCi markets Dirac-3 as an Entropy Quantum Computer; its hardware paper (Nguyen et al., *Communications Physics* 2025, arXiv:2407.04512) describes “a hybrid photonic-electronic computer that uses optical measurement and feedback to solve non-convex optimization problems”, and an independent critical comment (arXiv:2605.03612) argues its demonstrated claims do not beat state-of-the-art classical algorithms. We therefore treat the device as a photonic analog optimizer and claim no quantum resource. No gate-based experiment is claimed here. A Phase 2 Amazon Braket or Classiq arm will run against a matched classical baseline, and will be reported whatever the outcome.

The system. The production detector is a tuned gradient-boosted ensemble (XGBoost, LightGBM, CatBoost; equal 100-trial Optuna budgets). The candidate is CVQBoost: a pool of weak classifiers, one per feature subset, each casting a hard vote in $\{-1, +1\}$; ensemble weights are chosen by minimizing a quadratic objective on the simplex, which Dirac-3 solves natively as a quasi-continuous problem. The objective is quadratic in the weights for every configuration; “subset order” below refers to the number of features per weak learner (one, two or three), not to the polynomial degree of the Hamiltonian. The frozen configuration uses 91 learners over 13 features (13 singletons plus 78 correlation-ranked pairs); the largest hardware block uses 833 learners over 17 features at subset order 3. QCi documents 949 quasi-continuous variables for quadratic objectives; every configuration here is below that. A 100-variable account ceiling constrained most

of the campaign until a mid-campaign grant lifted it: a billing tier, not a device limit.

The protocol is the contribution. Before any run, we froze hypotheses, gates with numeric pass criteria, splits, seeds, metrics, statistical procedures and the hardware budget. Every change since is a dated amendment (32 to date) with rationale. Gates are scored as committed even when they fail; one gate (a 0.85 AUPRC floor for tuned XGBoost) failed at 0.8296, and we recorded the continuation as a deviation because we later found no peer-reviewed source for the floor. Every number carries an evidence tag ([HW] measured on Dirac-3, [SIM] simulated or classical, [PROJ] projected) and traces to a results record by configuration hash. Leakage controls run in code paths (split before preprocessing, train-only tuning, deduplication), and a shuffled-label control confirms the pipeline collapses to chance when signal is removed.

What we measured. ULB: 284,807 transactions, 1,081 exact duplicates removed, 60/20/20 stratified, ten seeds. IEEE-CIS: 590,540 transactions, rolling-origin evaluation by month. Intervals across seeds measure split dispersion, not sampling error, because the ten resplits share rows; directions therefore rest on seed unanimity. The minimum detectable effect (MDE) of 0.0268 AUPRC is the paired difference this design resolves at 80% power and alpha 0.05; it is a design parameter, not a significance boundary.

Result	Value	Tag
Tuned CatBoost, all features, ULB AUPRC (XGBoost, LightGBM within 0.013)	0.8368, CI 0.8150 to 0.8585	SIM
Shuffled-label control (base rate 0.0017)	0.0023	SIM
H1b, confirmatory: CVQBoost on Dirac-3 (0.7671) minus matched-13-feature CatBoost (0.8070)	-0.0399, CI -0.0571 to -0.0227; trails 9 of 10 seeds	HW
Hardware minus exact classical solve, same objective (subset order 2)	-0.0010 AUPRC; weight cosine 0.975 to 0.999	HW
Frozen pool: solved weights vs uniform weights	0.7687 vs 0.7659; tie-block reordering spans 0.0120	SIM
Fit-time class weighting, matched 6-feature pools	+0.0319, 10 of 10 seeds, above MDE	SIM
833-variable arm (17 features, order 3) vs matched 91-variable arm	+0.0256, CI +0.0203 to +0.0310, 10 of 10; absolute 0.7928	HW
Fraud caught at 0.1% review budget, CatBoost vs CVQBoost (0.5%: 85.5% vs 81.9%)	59.3% vs 56.5%	SIM
IEEE-CIS rolling-origin AUC-ROC, LightGBM (statement cites 0.9459, a leaderboard figure)	0.9139	SIM
IEEE-CIS feature ladder at k=17 (136 pair learners), hardware vs classical proxy (max cell gap 0.0013 over 12 cells)	0.1430 vs 0.1427	HW
Dirac-3 campaign	61 fits, 1,141 metered seconds, zero failures, zero retries	HW

Why the null, measured rather than argued. The CVQBoost objective is a convex quadratic on the simplex, so its optimum is unique and reachable in polynomial time; our classical solve is certified by KKT residual below $1e-9$. On the frozen pool the optimum is uniform to within $7.4e-06$ in L1 on every seed. The cause is measured at the pool: the trees are unbounded, 80 to 84 of the 91 learners reproduce the training labels exactly, and the Gram matrix is rank-one to numerical precision (second eigenvalue $1.0e-06$ of the largest, though nine exceed an absolute $1e-4$). The 0.0028 gap between solved and uniform weights is smaller than the 0.0120 that reordering the 95% tied block alone can produce, so it is tie-breaking, not signal. Neither a quantum nor a classical solver does useful work on that configuration. Hardware agrees with the exact solve (-0.0010 AUPRC; its objective is always above the exact minimum, by 0.013% to 0.413%), so the agreement is a solver-fidelity result, not a performance one. There is no speed argument at this size: a fit bills 4 to 5 device seconds against milliseconds for the exact solve.

Where the value is. Rebuilding the pool with four learner families and fit-time class weighting gives real diversity (Gram off-diagonal ratio 0.92 against 0.9988). The +0.0319 gain decomposes, against the same

per-seed comparator, into family diversity -0.0047 (4 of 10 seeds), fit-time class weighting +0.0328 (10 of 10), and shallower trees with distance-weighted neighbours +0.0038 (8 of 10). The optimizer’s own contribution is +0.0047 (SD 0.0028), below the MDE. The gain came from better weak learners, not better optimization. The 833-variable arm is the only positive result at scale, and two caveats travel with it: feature count and subset order moved together, so the gain is confounded, and the device solves the order-3 problem less faithfully (weight cosine 0.83 against 0.98). It trails full-feature CatBoost by 0.0440. The lower fidelity has a documented cause: QCi states an effective analog resolution of about 200:1, so with the sum constraint at 1 a diffuse optimum over 833 learners (about 0.0012 per weight) is below what the device can represent; every one of the eleven fits returned exact zeros and sparser weights than the exact solve. A device that cannot represent diffuse weight over more than about two hundred learners is a device whose native problem is sparse selection, which is the Phase 2 formulation in section 5.

Closest prior work. Loke et al. (“Improving Credit Card Transaction Fraud Detection Using CVQBoosting”, ICAART 2026, DOI 10.5220/0014628400004052) ran CVQBoost on the same hardware and benchmark and report mean AUC-PR above 0.8 (their best single-family arm, k-nearest neighbours, 0.8108). Their protocol is ten random 70/30 splits with SMOTE on the training fold, the regularization coefficient at zero (where our analysis shows the continuous optimum is not unique), and a semi-supervised anomaly detector (CAD, 0.7423) as comparator, so it is not like-for-like with our tuned gradient-boosted baseline; it is a pre-registered replication target (H1a), not an explanation of our null. Emami et al. (arXiv:2503.11273) report AUC parity with XGBoost on ULB only after training-fold rebalancing (XGBoost leads in 23 of 24 cells; CVQBoost’s single win is ADASYN at a 1:1 ratio) and a training-runtime advantage on synthetic data from 1M to 70M samples. They also report that Dirac-3 solution energies fall slightly behind Hexaly and SLSQP at 500 to 900 variables, which they attribute to Hamiltonians exceeding the device’s 23 dB dynamic-range limit; that is consistent with the sparsified solutions we observe at 833 variables (weight cosine 0.83). Our null is consistent with their accuracy result and is measured under a stricter condition: AUPRC at 0.17% prevalence, without resampling.

3. Feasibility and resource requirements

Data. Phase 1 used two public benchmarks, checksummed in the repository manifest and not redistributed. Phase 2 needs what no public benchmark carries: merchant, device and account-age features, and labels with a chargeback maturity window. That is a bank-side dependency; the trial design in section 4 works on whatever subset a partner can expose.

Quantum compute. Dirac-3 access is a separate QCi allocation: 3,000 seconds granted 2026-09-09, of which 1,961 remain. Costs are measured, not estimated: about 82 seconds per fit at 833 variables (maximum 92), 4 to 9 at 136, 4 to 5 at 91. A Phase 2 campaign at the scales in section 5 is a few thousand seconds, affordable within one further grant of the same size. Schedule: all six experiments in section 5 complete within the first two months of the sprint. Month 1: experiments 1, 2 and 4 (zero device seconds; proxy and rolling-origin folds). Month 2: experiment 3 on the integer path, its device cost fixed by one probe fit before the block is committed, then experiments 5 and 6. Months 3 and 4 are reserved for the temporal protocol on partner data if a bank-side feature set is exposed, external review, and reporting. We will use the challenge’s AWS credits and Classiq platform to fund a gate-based arm, and will run and report it whatever the outcome.

Device limits that shape the design. Two bind: a sum-over-levels cap puts binary selection near 474 variables, and the 200:1 resolution limits a solve to about 200 learners. Section 5 therefore selects within blocks of 200, pruning learners carrying no weight and grouping those whose features interact, rather than posing 833 at once.

Classical compute and software. Everything except metered solves runs on one workstation. The bottleneck is pool construction, not device time: about 12 minutes of CPU per fit at 833 variables, with Hamiltonian construction growing with the square of pool size. eqc-models 0.21.0 and qci-client 5.0.2, pinned in a locked environment; the pool build requires POSIX fork (Linux).

Assumptions and constraints stated. Real-time scoring latency (low tens of milliseconds) is unmeasured for both arms; CVQBoost inference is a linear vote, so we expect it to be cheap, but p50/p95/p99 on named

hardware is Phase 2 work. Calibration is specified for the trial, not fitted. Fairness testing across protected characteristics is absent, because the benchmarks’ features are anonymized principal components; it becomes a first-class requirement the moment real customer data is in scope (Equality Act 2010; FCA FG22/5 para 5.12 names algorithms that produce systematically worse outcomes for some groups as an example of not acting in good faith, and chapter 11 expects firms to monitor outcomes by customer group). Label latency: chargeback labels arrive weeks late, so evaluation windows must treat recent unlabeled transactions as censored. Throughput: a solo entry; the defects that mattered in Phase 1 were found by adversarial review, not by the test suite, so external review is budgeted as a Phase 2 line item.

4. Expected impact

What a successful PoC would show. Either (a) a quantum solver reaching a certified-optimal or better solution to the cardinality-constrained selection problem at lower end-to-end cost than a time-capped MIQP solver, or (b) a router combining the classical champion and the quantum-inspired specialist that raises fraud caught at the 0.5% review budget by at least one percentage point above the classical champion alone, on out-of-time folds. The measured gap at that budget is 3.6 points, so the target is to close it and add one. Neither is assumed.

The arithmetic of the metric [PROJ]. For a portfolio of one million transactions a day at 0.17% fraud, a 0.1% review budget funds 1,000 alerts against about 1,700 frauds, capping attainable recall at 58.8% before any model is chosen. Within that ceiling each percentage point of recall is about 17 frauds a day. The 2.8-point gap we measure between the production detector and the current quantum-inspired arm is therefore about 48 frauds a day at that scale, which is the quantity a trial must move. This illustrates the arithmetic; it assumes a portfolio we have not seen, and the benchmark spans two days. On our test fold the 0.1% budget is saturated (CatBoost reaches 98.8% of the ceiling), so the 0.5% budget, where the gap is 3.6 points, is the informative column.

Explainability, which is the bar that gates adoption. A CVQBoost score is an exact sum of per-learner terms, each naming its one or two features and its weight; we measured this on a real flagged fraud. At the current configuration the explanation is exact but long (46 of 91 learners reach half the total weight); the sparse formulation in section 5 would shorten it. On anonymized features this is an audit decomposition, not a customer-meaningful explanation; that gap closes only on named bank features.

Production trial, 90 days. Days 1 to 30: the tuned classical ensemble scores live traffic in shadow, compared on fraud caught within the standing review budget. Days 31 to 60: champion/challenger on a small routed share with a rollback trigger on precision at budget. Days 61 to 90: the quantum-inspired arm enters the shadow track as a specialist on segments identified in Phase 2. Retraining triggers are population stability and precision-at-budget degradation, not a calendar. The champion is not deployable as fitted: the benchmark’s PCA transform is unpublished, so onboarding needs feature mapping, retraining, temporal validation, calibration and governance approval.

5. Validation plan

Already scored. The confirmatory endpoint H1b (CVQBoost on Dirac-3 vs best matched-feature GBDT, ULB AUPRC, ten seeds) is null at -0.0399 [HW]. Everything else in this proposal is exploratory and labelled so.

Phase 2 experiments, in order, each with its acceptance criterion written before it runs.

1. *Isolate the confound (first, zero device seconds).* The 153-variable ULB cell at $k=17$ and subset order 2 (17 singles plus 136 pairs) separates feature count from subset order in the +0.0256 result. Runs on the classical proxy. Until it runs, the gain cannot be attributed.
2. *Temporal validity.* On the single time-ordered ULB split, the configuration our preregistered rule selected fell from 0.7671 to 0.7095 AUPRC while the rejected 9-feature configuration rose from 0.7014 to 0.7776 [HW]. One split proves nothing, but it points the wrong way for our own choice; IEEE-CIS rolling-origin folds supply the temporal test that ULB’s two-day span cannot. No configuration of this arm is recommended for a production decision path until this is settled.

3. *The formulation where the optimizer must work.* Choosing the best sparse subset of weak learners under a cardinality constraint is NP-hard in general and is the native problem class of Dirac-3’s integer solver. NP-hardness bears on worst-case instances and does not establish that a quantum device is necessary or advantageous on the finite instances we would run. Controls: a time-capped commercial MIQP solve (which may still certify a global optimum at these sizes), greedy selection, and simulated annealing. A win against a certified-optimal classical solve is a result; a win against no control is not.
4. *Replication under source protocol.* H1a reproduces Loke et al.’s configuration under their split and comparator, target AUC-PR 0.80, then re-evaluates it under ours.
5. *Segment transfer (H5).* Replicate a prior Dirac-3 campaign’s in-segment wins (single seed, small margins) with seeds and intervals, then test transfer to fraud segments against a matched random-segment control.
6. *Scaling claim.* End-to-end training time against sample count, including Hamiltonian construction, which decides whether the vendor’s runtime claim survives on this problem.

Metrics. Primary: fraud caught within a fixed review budget (0.1% and 0.5%) on out-of-time folds. Secondary: AUPRC, precision at budget, hardware-vs-proxy fidelity (AUPRC gap, weight cosine, objective gap), metered device seconds per fit, and end-to-end wall time. Acceptance for the router: at least one point of recall at the 0.5% budget above the classical champion alone, out of time, on prespecified segments against a matched random-segment control. Acceptance for any hardware claim: lower end-to-end cost at matched solution quality, or better prespecified predictive performance under a matched protocol. Hardware cannot beat a certified global optimum of the identical convex objective, so “the device found a better answer” is not an available claim on the continuous formulation and we will not make it.

What retires the approach. The preregistration names three conditions that together retire it: a null on H1b (fired), a non-positive feature-ladder slope (fired at -0.006 AUPRC per feature; the classical arm improves faster as features are added), and a failure of H5 segment transfer (unrun). If H5 also fails, the honest conclusion is that this formulation does not belong in a fraud stack, and we will say so. Separately, quantum feature engineering with order-matched classical twins moved the quantum-minus-classical difference by -0.0115 against a paired SD of 0.0135, below the MDE, so representation is not the lever [SIM]. The classical bar follows the complete-lineup principle of Mancilla and Tagliani (arXiv:2607.15815), re-implemented independently here: every representation cell carries a trained-frequency GAM, a GA2M and an order-matched joint twin alongside the three GBDTs; the GAM twin scores 0.7893 against the quantum arm’s 0.7646, and omitting such twins is how apparent quantum wins are manufactured.

How the evidence stays checkable. Every number originates in a results record with a configuration hash and evidence tag, regenerated by a scoring script; full tables, the gate register and references are in Appendices A to C. Across the twelve hardware ladder cells the errors scatter in both directions (five above the classical arm, five below, two exact) against a comparator certified by KKT residual. An earlier hardware ladder that appeared to triple CVQBoost’s AUPRC was traced the same day to a pool-size mismatch, withdrawn and re-run; the superseded figures appear nowhere in this submission. Sixty QCi job identifiers, 48 raw device responses and the job parameters (relaxation_schedule 2, sum_constraint 1, num_samples 8) are version-controlled. The preregistration, the gate report, all 32 amendments (including three corrections made after figures were published and one correction that was itself wrong), per-seed values and the worked explainability decomposition are in the public repository, frozen at commit 95751b9 (tag `prereg-freeze`): <https://github.com/kimmeyh/hsbc-quantum-fraud-2026>.

6. Hybrid integration

The deployed shape is a classical detector in the transaction path with a quantum-inspired specialist behind a router, a route-and-blend deployment in the sense of arXiv:2607.15815, with the router gated on features independent of the specialist’s. The classical ensemble scores every transaction. The specialist is invoked only on segments where it is measured to help; its inference is a linear vote, so the quantum device sits entirely at training and model-selection time, outside the live path, and nothing in production depends on a quantum device being reachable. The rationale is measured, not assumed: on every configuration tested the classical ensemble is the champion, so the specialist must earn its place on out-of-time segment lift, and the router is a Phase 2 hypothesis with an acceptance criterion, not a design we assert.

7. Team capability

Solo entry led by Harold Kimmey, with Claude Code disclosed as an AI development tool used under the team lead's direction. Measured relevant experience: prior Dirac-3 campaigns with this model family, including the configuration that seeded this work; quantum machine learning explainability; hands-on Amazon Braket, which makes a Phase 2 gate-based arm a commitment. The Phase 1 apparatus, the hardware campaign and this proposal were built in under three weeks. Throughput binds a solo entry, and Phase 2 is scoped to it: six experiments in priority order, the first costing zero device seconds, with external review budgeted.